

NMR Spectroscopy

Environments, shift, integration and splitting

Name _____ Class _____ Date _____

¹H NMR DATA

δ (ppm): alkyl 0.9–1.7 · next to C=O 2.0–2.7 · on C–O 3.3–4.3 · alkene 4.5–6.5 · aromatic 6.5–8.0 · CHO 9–10 · COOH 10–12 · OH 1–5

δ (ppm) = shift from TMS (Hz) $\div$ spectrometer frequency (MHz) · n + 1 rule: n neighbouring H give n + 1 peaks

¹³C (ppm): alkyl 0–50 · C–O 50–90 · alkene/aromatic 110–160 · C=O 160–220 · IR: C=O 1680–1750 cm⁻¹

Draw the structure and label each H environment. Show units in every calculation step.

A. Environments

1. List the four kinds of information a ¹H NMR spectrum provides.

2. Give the number of ¹H NMR signals and the ratio of their areas for: (a) ethane (b) propane (c) 2-methylpropane, (CH₃)₃CH (d) benzene (e) methyl propanoate, CH₃CH₂COOCH₃

B. Chemical shift

3. (a) A signal is 2220 Hz from TMS on a 600 MHz spectrometer. Calculate δ . (b) How many Hz from TMS is a signal at 1.20 ppm on a 300 MHz spectrometer?

C. Splitting and predicting spectra

4. Predict the splitting of each signal in: (a) chloroethane, $\text{CH}_3\text{CH}_2\text{Cl}$ (b) 2-bromopropane, $(\text{CH}_3)_2\text{CHBr}$

5. Predict the full ^1H NMR spectrum (δ , area, splitting) of propanoic acid, $\text{CH}_3\text{CH}_2\text{COOH}$.

6. How many ^{13}C NMR signals do these give: (a) propanone (b) ethyl ethanoate (c) benzene?

D. Solving structures

7. A compound $\text{C}_3\text{H}_8\text{O}$ shows: doublet (6H) at 1.2 ppm, septet (1H) at 4.0 ppm, singlet (1H) at 2.2 ppm. Identify it and assign each signal.

E. Challenge

8. A compound $\text{C}_4\text{H}_8\text{O}$ has a strong IR absorption at 1715 cm^{-1} and a molecular ion at m/z 72. Its ^1H NMR spectrum shows: triplet (3H) at 1.1 ppm, singlet (3H) at 2.1 ppm, quartet (2H) at 2.4 ppm. Identify the compound and explain each piece of evidence.